

Storage Device

Chapter Five

Storage devices

- Storage devices are used to store data permanently.
- Data is stored on them for future use.

We have different storage media:

- Floppy disk
 - Hard Disk
 - Compact Disc (CD)
 - Flash Disk
- Each of these devices have a device that **read** data from them or **write** data onto them. This device is called **drive**

Hard Disk

- Hard disk is the data center of PC.
- It is here that all your programs and data are stored.
- A hard disk drive contains rigid, disk-shaped platters.
- The platters are coated by magnetic substance on both sides.
- Data is written to hard disk by magnetizing and demagnetizing the surface of the platters.
- Data is stored on both sides of the platter.

Types of Hard Disks

Based on the **connectors** used to connect hard disks to motherboard, there are three types of hard disks:

- **SCSI** (Small Computer System Interface)
- **SATA** (Serial ATA(*Advanced Technology Attachment*)).
- **IDE** (Integrated Drive Electronics) or ATA or PATA(Parallel ATA)

IDE...

1) IDE

It is also called **ATA**.

- In IDE, the interface electronics or controller is built into the drive itself; why it is called Integrated Drive Electronics.
- Today, **IDE** is used to connect not only hard disk, but also CD-ROM drives, tape drives, and high capacity super Disk floppy drives.
- **Bandwidth**: 16 MB/s originally; later 33, 66, 100 and 133 MB/s
- **Max devices**: 2 (one as **master** and the other as **slave**)
- **Connector**: Parallel Cable 40 or 80 wires

Cable Select

- A drive setting called *cable select* .
- A drive set to "cable select" automatically configures itself as master or slave, according to its **position** on the cable.
- Cable select is controlled by pin 28.
- The host adapter grounds this pin; if a device sees that the pin is grounded, it becomes the **master** device; if it sees that pin 28 is **open**, the device becomes the slave device.
- This setting is usually chosen by placing a jumper on the "cable select" position, usually marked CS, rather than on the "master" or "slave" position.
- Note, pin 28 is only used by the drive to determine its position on the cable and thus whether it's the "master" or "slave"; it isn't in any way a "drive select" pin used when the host is issuing commands so if a drive is jumpered to "master" or "slave" its position on the cable doesn't have to match the pin 28 meaning of "master" and "slave".

2. SATA(Serial Advanced Technology Attachment)

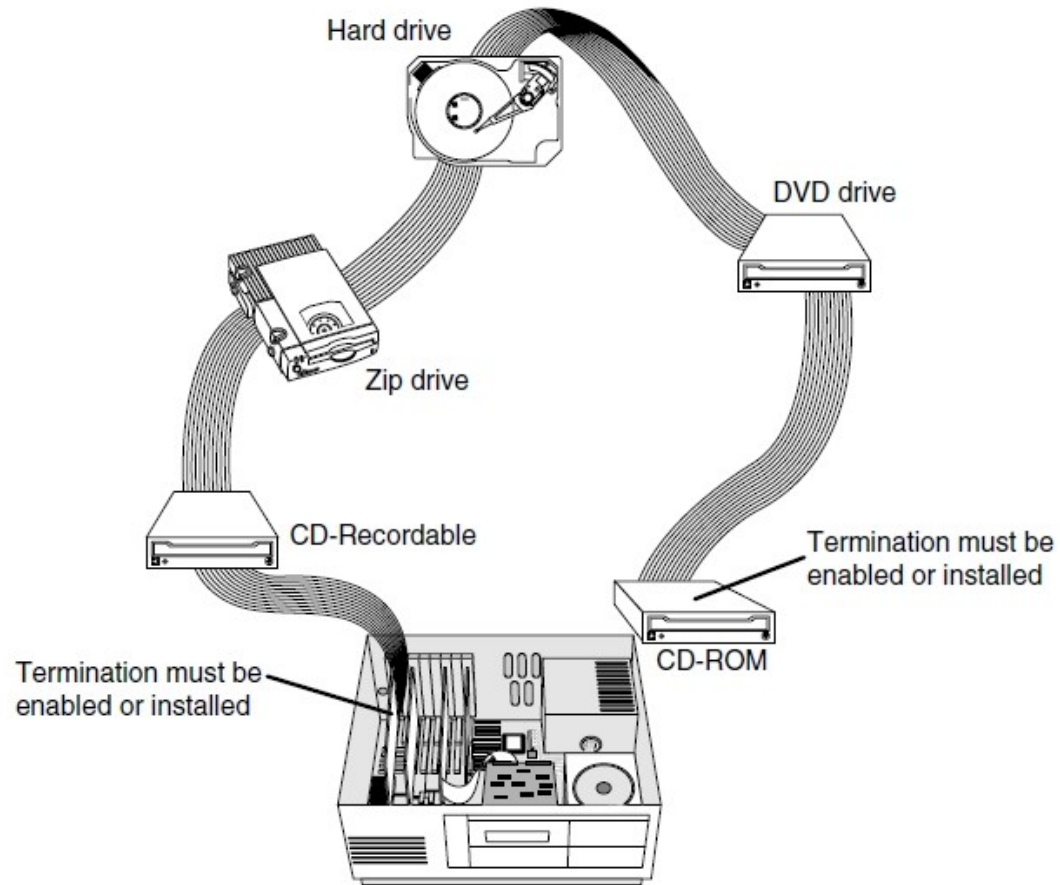
- To get **faster** speed (and if your hardware supports it (hard drive and system board), you can use SATA, or Serial ATA, which will give you much faster speeds for hard drive access.
- SATA uses special cables that are NOT compatible with standard ATA hard drives.
- The main advantages over the older parallel ATA interface are **faster** data transfer, ability to remove or add devices while operating (hot swapping), thinner cables that let air cooling work more efficiently, and more reliable operation with tighter data integrity checks.
- ATA adapters and devices communicate over a high-speed serial cable.



3) SCSI

- The Small Computer System Interface (SCSI, pronounced as scuzzy) standard.
- It is a technology which allows connecting different hardware devices such as hard disk drives, CD-ROM drives, tape streamers and scanners to a single adapter.
- Most systems support up to 4 host adapters, each with up to 15 devices.
- SCSI is a high end technology.
- It is a fast interface generally suited to high performance machines.
- It is especially used in high end PC's such as network servers, workstations, or anywhere where storage performance is needed.

SCSI...



HDD Passwords and Security



Chapter 6

The Power Supply

Power Supplier

The basic function of power supply is

- Supplying power to the system,
- **Convert** the type of electrical power available (Alternate Current-AC) to the type computer circuitry can use (Direct Current).
- Ensures that the system does not run unless power supply is **sufficient** to operate the system **properly** (the power supply actually prevents the computer from running (starting up) or operating until all correct power levels are present, Each power supply completes **internal checks** and tests before allowing the system to *start*. If tests are successful, the power supply sends a special signal called Power Good to the motherboard. If this signal not present, computer does not run.)

Power Supplier....



Power Supplier...

- **Fact:** Almost everything in your home runs off standard AC power. But they actually use DC internally.
- E.g printers, external modem, scanner, etc.
- **Attention:** plugging the power connectors in the wrong direction could damage the computer, Pay attention when you plug.

Connectors to Peripheral Hardware

- Two standard types of connectors are used to connect power to peripheral devices:
 - Molex connectors
 - Mini connectors

Connectors to Peripheral Hardware...

1. Molex Connector

- This is the most commonly used connector.
- Hard disk drives, internal tape drives, CD-ROM drives, DVD drives, and older 5^{1/4} floppy drives use this power connector. It has four pins.

2. Mini Connector

- It is used primarily for 3.5 floppy disk drive.
- It has four pins and four wires.
- Connecting the connector backward could damage the system.
- The good thing is it does not fit in the wrong direction

Power Failure

- Power supplies are affected by **quality** of local power source.

Common power delivery problems are:

- Spikes
- Surges
- Sags
- Brown outs & black outs

Power Failure...

1. Surge

- A brief increase in the voltage source .

2. Spikes

- Are very short over voltage conditions.

3. Sags

- A brief decrease of voltage at power source.

4. Brown Outs

- If a sag lasts longer than 1 second, it is called brown out.

5. Black Outs

- This is a complete power failure.
- This could be caused by equipment failure, or accidental cutting of power cable.

Power Control

There are different devices that enable us to control power.

a) UPS (Uninterruptible Power Supply):

- A UPS helps in black out situations when the power is gone totally and brown outs, when voltage is low though there is power.
- UPS has batteries that provides back up power. When the power is gone, the battery provides voltage to the PC so that you can save your work. But it does not provide **unlimited** power.

Power Control...

There are two types of UPS:

1. Standby UPS:

- SPS has a battery that begins generating power as soon as the unit detects a sag(fall) in the power supply.
- It takes only a split of seconds to come online.
- The disadvantage is that till the UPS become online, your data might get lost.

2. Online UPS:

- It provides electricity to the PC all the time. It uses electricity from the AC outlet to simply recharge its batteries.
- When the power goes, the data is not affected because the UPS is supplying power.

Power Control...



Fig UPS (front and back view) and fuse

Power Control...

b) Surge Suppressor:

- help to absorb power surges(flows), so that your computer does not feel their effects.



- Fig Surge suppressor

Power Control...

c) Voltage Stabilizer

- It is a transformer that **delivers** relatively **constant** output when output voltage changes over time.
- The output voltage is regulated using transistor.

Battery

- Computer has a Chip that combines real time clock and non-volatile memory.
- This chip is the CMOS chip (CMOS RAM).
- They are designed to consume low power.

Battery

- The CMOS chip stores basic system configuration like amount of memory installed, type of floppy and hard disk drives, type of keyboard and mouse, etc.
- This is powered by battery when system is off.
- This battery keeps the time chip running and the CMOS powered.
- Most systems use lithium type battery because they have very long life.
- Most of the higher quality systems sold today have a new type of chip that has the battery embedded in it. These chips are notable for long life. Under normal conditions, the battery lasts for **10** years. Most of the time, these chips-battery combinations are installed in a socket on the motherboard. Some systems do not use battery at all.

Troubleshoot Power Supply

PC problems that often relate to PSU are:

- Any power-on or system startup failures or lockups
- Spontaneous rebooting or intermittent(irregular) lockups during normal operation
- Intermittent parity (uniform) check or other memory-type errors
- Hard disk and fan simultaneously failing to spin
- Overheating due to fan failure
- Small brownouts that cause the system to reset
- Electric shocks/wave felt on the system case or connectors
- Slight/small static discharges that disrupt system operation